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个人简介

孙宝发，南开大学生命科学学院研究员。中国科学院动物研究所博士毕业，2013-2021年在中国科学院北京基因组研究所担任助理研究员和副研究员，入选中科院青促会会员，天津市青年科技人才，科睿唯安2023-2024年度“全球高被引科学家”。主要从事神经精神疾病，心脑血管疾病及多种肿瘤的单细胞时空多组学生物信息学分析及功能研究。在国际主流杂志Nature, Nature Immunology, Science Immunology, Nature Cell Biology, Nature Communications, Science Translational Medicine, Molecular Cell, Cell Research等杂志发表论文70多篇，一作和通讯作者文章总影响因子达600多，引用近10000次。主持/参与过10余项国家自然科学基金。本课题组长期招聘信息及实验博士后，有意者请联系呀，等你哦！欢迎志同道合小朋友大朋友一起玩耍，玩它个天翻地覆，玩出个大好前途！

教育经历

2002.09 -2006.06 生物科学学士，湖南科技大学；（其实是师范生，学校合并前叫湘潭师院）2006.09 -2009.06 生物化学与分子生物学硕士，汕头大学，导师：胡忠；（想回北方，调剂的更远了。。）2009.09 -2013.01 动物学博士，中国科学院大学动物研究所，导师：黄大卫；（没赶上08奥运会，遗憾呀）

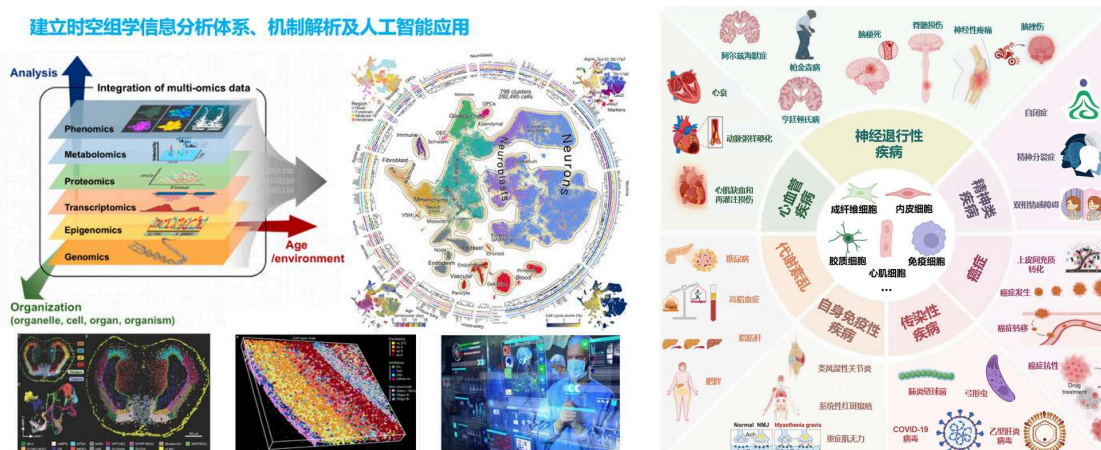
工作经历

2013.07-2015.12 助理研究员，中国科学院北京基因组研究所；2016.01-2021.04 副研究员，中国科学院北京基因组研究所；2021.05-至今 研究员 南开大学生命科学学院，博导

研究方向

以免疫为核心的重大疾病生物信息与人工智能研究组

（神经、心脑血管疾病及肿瘤单细胞时空多组学，表观组和人工智能应用）

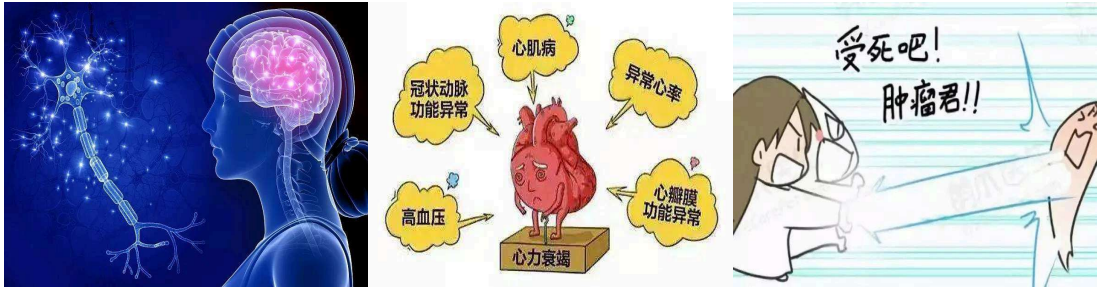


课题组研究方向以免疫为核心，集中于与神经疾病、心脑血管疾病、肿瘤及发育相关的单细胞及表观组学等测序技术及分析方法开发、调控机制解析及人工智能应用。

课题组旨在以解决与多种疾病的生物医药相关基础科学问题为导向，在技术开发、机制解析基础上，整合多维组学及临床医学信息对神经疾病、心脑血管疾病及肿瘤通过人工智能进行精准预测、诊断和治疗，实现智慧医疗。

内容：通过整合多维组学信息解析神经疾病、肿瘤、免疫及发育等功能调控。

- 1. 通过单细胞及空间转录组揭示神经疾病、肿瘤发生、转移、耐药、免疫等分子机理。
- 2. 揭示表观基因组及表观转录组学在多种疾病（神经疾病、心脑血管疾病及肿瘤）及生物学过程的调控作用。
- 3. 在单细胞时空层面阐明RNA结合蛋白及RNA加工对各种生理及病理的作用。
- 4. 明确相分离在多种疾病（神经疾病、心脑血管疾病及肿瘤）及免疫调控中的作用机理。



都是发CNS热门方向。
申明：不是为了蹭热点，单纯为了做科研！（迷之凡尔赛，你懂的）。
本课题组长期招聘生物信息及功能实验博士后及联培生，有意者请随时联系！

一个本本闯天下



仗剑走天涯



持本闯天下

科研成果



发现RNA修饰甲基转移酶新组分以及新结合蛋白，阐明RNA修饰调控基因加工及对发育、再生和多种疾病调控的分子机理。
通过单细胞多组学揭示神经疾病、心脑血管疾病、病毒感染、癌症等疾病发生的分子机制。

一、发表论文

第一（共同第一）作者(#); 通讯（共同通讯）作者(*)

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- 2) Yihao Liu#, Jiayu Lin#, Zhengwei Yu#, Xueying Li#, Xueqi Lv#, Pengyi Liu#, Xiuqiao Sun, Zhen Zhang, Xia Gao, Keyan Sun, Dan Li, Jingfeng Li, Yang Liu, Yu Jiang, Siyi Zou, Jianping Lin, **Baofa Sun***, Da Fu*, Baiyong Shen*. Tumor-associated Schwann cell remodeling under metabolic stress via lactate sensing orchestrates pancreatic ductal adenocarcinoma development. *Cell Metabolism*. S1550-4131(25)00335-3 (2025).
- 3) Shoubao Ma#, **Baofa Sun#**, Songqi Duan#, Jingjing Han#, Tasha Barr, Jianying Zhang, Marc B. Bissonnette, Marcin Kortylewski, Chuan He, Jianjun Chen , Michael A. Caligiuri*, Jianhua Yu*. 2) YTHDF2 orchestrates tumor-associated macrophage reprogramming and controls antitumor immunity through CD8(+) T cells. *Nature Immunology* 24(2): 255-266 (2023).
- 4) Sai Xiao#, Shoubao Ma#, **Baofa Sun#**, Wenchen Pu, Songqi Duan, Jingjing Han, Yaqun Hong, Jianying Zhang, Yong Peng, Chuan He, Ping Yi*, Michael A. Caligiuri*, Jianhua Yu*. The tumor-intrinsic role of the m6A reader YTHDF2 in regulating immune evasion. *Science Immunology* 9(95): eadl2171 (2024).
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- 7) Qianwen Liu#, Jingfeng Li#, Xiuqiao Sun#, Jiayu Lin, Zhengwei Yu, Yue Xiao, Dan Li, **Baofa Sun***, Haili Bao* and Yihao Liu*. Immunosenescence and cancer: molecular hallmarks, tumor microenvironment remodeling, and age-specific immunotherapy challenges. *Journal of Hematology & Oncology* 18(1):81 (2025).
- 8) Yihao Liu#, Minmin Shi#, Xingfeng He#, Yizhi Cao, Pengyi Liu, Fanlu Li, Siyi Zou, Chenlei Wen, Qian Zhan, Zhiwei Xu, Jiancheng Wang*, **Baofa Sun*** and Baiyong Shen*. LncRNA-PACERR induces pro-tumour macrophages via interacting with miR-671-3p and m6A-reader IGF2BP2 in pancreatic ductal adenocarcinoma. *Journal of Hematology & Oncology* 15(1):52 (2022).
- 9) Yaqian Li#, Qingyang Zhang#, **Baofa Sun#**, Yidi Ma, Ye Zhang, Min Wang, Congcong Ma, Honghui Shi, Zhijing Sun, Juan Chen, Yun-Gui Yang*, Lan Zhu*. Single-cell transcriptome profiling of the vaginal wall in women with severe anterior vaginal prolapse, *Nature Communications* 12(1): 87 (2021).
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1. 杨运桂, 杨莹, 孙宝发, 杨鑫, 孙慧颖, 孙敏, 张兵, 黄春敏, 2018年2月6日, ZL 201510557083.9, RNA 5mC重亚硫酸盐测序的文库构建方法及其应用

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2018 发表在Nature杂志的 “m⁶A modulates haematopoietic stem and progenitor cell specification” 一文被评为2017年度 “中国生命科学领域十大进展”。

2018 发表在Cell Research上的题为“5-methylcytosine promotes mRNA export - NSUN2 as the methyltransferase and ALYREF as an m⁵C reader”的文章获得了2017年赛诺菲-Cell Research杰出研究文章奖。

2018 “5-methylcytosine promotes mRNA export - NSUN2 as the methyltransferase and ALYREF as an m⁵C reader” 和 “Mammalian WTAP is a regulatory subunit of the RNA N6-methyladenosine methyltransferase” 被选为第三届中国科协优秀科技论文。

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